Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/11/2004
PI Name:	Townsend, Lawrence W. Ph.D.		
Project Title:	Radiation Transport Code Development for Space Radiation Shielding Applications		
Division Name:	Physical Sciences		
Program/Discipline:	RADIATION SHIELDING		
Program/Discipline--Element/Subdiscipline:	RADIATION SHIELDING--Radiation shielding		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	ltownsen@tennessee.edu	**Fax:**	FY 865-974-0668
PI Organization Type:	UNIVERSITY	**Phone:**	865-974-7569
Organization Name:	University of Tennessee		
PI Address 1:	Nuclear Engineering Department		
PI Address 2:	University of Tennessee		
PI Web Page:	Aspects of the FLUKA code development work are available at http://fleur.cern.ch/		
City:	Knoxville	**State:**	TN
Zip Code:	37996-2300	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	12/01/2002	**End Date:**	11/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	10	**No. of Master' Degrees:**	1
No. of Master's Candidates:	4	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Barghouty, N	**Contact Phone:**	256 544-0238
Contact Email:	Nasser.Barghouty@msfc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Pinsky, L. S. (University of Houston) Empl, A. (University of Houston) Ferrari, A. (CERN & INFN) Cerutti, F. (INFN) Garzelli, M.-V. (INFN) Ranft, J. (CERN & Univ. of Leipzig (Germany)) Gabriel, T. (Scientific Inv. & Develop.) Sihver, L. (Roanoke College/Chalmers University) Adams, J. H. (NASA MSFC) Watts, J. (NASA MSFC) Wilson, T. L. (NASA JSC)		
Grant/Contract No.:	NAG8-1901		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The task, to be carried out by a consortium of researchers from the University of Tennessee (lead institution), the University of Houston, Roanoke College and various government and national laboratories, involves the development of a standard set of radiation transport computer codes for NASA applications. This effort involves further improvements of the Monte Carlo codes HETC and FLUKA. Included in these improvements is the development of nuclear reaction databases necessary to extend these codes to heavy ion transport. Also planned are analyses of the remaining deficiencies in these codes and in the deterministic code HZETRN, and the determination of what is needed to develop a standard NASA radiation shielding code. Error and sensitivity analyses will be performed to determine errors induced in calculated particle spectra by uncertainties in fragmentation cross sections. The improved codes will be validated by comparisons with measured transport data as they become available. Methods of solving the Boltzmann equation for 3-D radiation transport will also be explored using the numerical method of lines approach. Finally, techniques to facilitate the geometric modeling of complex and detailed 3-D configurations such as spacecraft for use with these transport codes will be explored. To develop a standard space radiation transport code/set of codes for NASA applications. This goal will be accomplished by further development of the HZETRN, HETC and FLUKA radiation transport codes. The ability to simulate the radiation environment in space and the transport of these particles through spacecraft shielding materials is crucial to concerns such as crew health and electronic equipment performance. The need exists to understand both the actual measurement of crew radiation doses recorded on current missions as well as to explore the predicted doses for future missions and new spacecraft designs. These planning tasks call for significant flexibility to simulate a wide variety of conditions such as those likely to exist within spacecraft in interplanetary space, and on the surface of the Moon and Mars. In addition, many experiments need to simulate the effects that space radiation has on their instruments. These effects can either be direct as part of the actual measurements or as a background to it. Further, there can be operational concerns such as degradation of instruments from radiation damage or data interference as in the case of Single Event Upsets. The improvement in the accuracy of the software that NASA currently uses for these tasks that will result from this project will substantially increase the reliability of the projected results.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	1. The HETC radiation transport code has been modified to enable it to be used to simulate the transport of high-energy heavy ion radiation fields through shield materials and the human body. The new code, called HETC-HEDS, is now capable of carrying out a full 3D transport of the complete space radiation. These modifications incorporated in HETC include all physics necessary to transport heavy ions and to describe their nuclear interactions with the target atoms and nuclei. The details of the nuclear interactions between the incident high-energy heavy ions (so-called HZE particles) are provided by a HZE particle event generator, which models the collisions between the transported fields and the target nuclei. The event generator decides what type of collision has occurred (elastic or inelastic) and for inelastic collisions, the secondary radiation particle types, momenta and energies, and directions of travel. Details of the modified version of HETC-HEDS and the HZE particle event generator have been published/accepted for publication in several peer-reviewed journal articles. 2. Validation of the newly-modified HETC-HEDS radiation transport code has begun by comparing predictions of fragment yields, from the nuclear interactions of high-energy heavy ion beams with thick targets, with measured fluences provided by the NASA-funded Measurements Consortium, using beams at the Brookhaven National Laboratory (AGS and NSRL). The initial comparisons have been reported at a major international conference and have been accepted for publication in a peer-reviewed journal. Additional comparisons with available data provided by the Measurements Consortium are in progress. Modifications to the HETC-HEDS code and the HZE particle event generator will be made, if necessary, from these comparisons with measured data. 3. For the FLUKA radiation transport code, the work continued during this second year of the project includes a continuation of the primary effort to develop a new event generator for nucleus-nucleus interactions for the energy region from 100 MeV/nucleon up through a few GeV/nucleon. Our primary efforts are based on an evolution of the basic rQMD (relativistic Quantum Molecular Dynamics) model as an improvement over the typical Intra-Nuclear Cascade (INC) based approaches used elsewhere (including by us in prior codes). We have also added the photo-nuclear disintegration cross sections into the event generators for nucleus-nucleus interactions. Versions of these new codes are being inserted into FLUKA for testing against known data. In addition, we are initiating an effort to modify a Nuclear Physics code based on the Boltzmann Master Equation (BME), to simulate interactions below 100 MeV/nucleon down to the threshold energies for inelastic reactions. Finally, we are upgrading the neutron transport capabilities to reflect the most recent publicly available neutron cross section data. 4. Beyond the development of these new physics models for FLUKA, we are also continuing to support an effort to improve the geometric modeling techniques used to represent the materials and geometries of the spacecraft, along with the tools to input this information, and eventually to convert it directly from CAD formats.
Bibliography Type:	Description: (Last Updated: 06/13/2007)
Articles in Peer-reviewed Journals	Miller, T. M.; Townsend, L. W. "Double Differential Light Ion Production Cross Sections" Radiation Protection Dosimetry , Vol. 110, Nos. 1-4, pp. 57-60 , Sep-2004
Articles in Peer-reviewed Journals	Miller, T. M.; Townsend, L. W. "Double Differential Heavy Ion Production Cross Sections" Radiation Protection Dosimetry , Vol 110, Nos. 1-4, pp. 53-56 , Sep-2004
Articles in Peer-reviewed Journals	Miller, T. M.; Townsend, L. W. "Comprehensive Cross Section Database Development for Generalized Three Dimensional Radiation Transport Codes" Nuclear Science and Engineering (in press). , Jan-2005
Articles in Peer-reviewed Journals	Townsend, L. W.; Miller, T. M.; Gabriel, T. A. "HETC Radiation Transport Code Development for Cosmic Ray Shielding Applications in Space" Radiation Protection Dosimetry (accepted for publication). , Aug-2004
Articles in Peer-reviewed Journals	Andersen, V.; Ballarini, F.; Battistoni, G.; Campanella, M.; Carboni, M.; Cerutti, F.; Empl, A.; Fassò, A.; Ferrari, A.; Gadioli, E.; Garzelli, M.V.; Lee, K.; Ottolenghi, A.; Pelliccioni, M.; Pinsky, L.S.; Ranft, J.; Roesler, G.; Sala, P.R.; Wilson, T.L. "The FLUKA code for space applications: recent developments" Adv. Space Res. (in press) , Sep-2004

Articles in Peer-reviewed Journals	Battistoni, G.; Cavinato, M.; Cerutti, F.; Clivio, A.; Fabrici, E.; Gadioli; E., Gadioli Erba, E.; Garzelli, M.V.; Mairani, A.; Empl, A.; Pinsky, L.S.; Ballarini, F.; Ottolenghi, A.; Fassò, A.; Ferrari, A.; Ranft, J.; Sala, P.R. "Heavy ion interactions from Coulomb barrier to few GeV/n: Boltzmann Master Equation theory and FLUKA code performances" Brazilian Journal of Physics (in press) , Sep-2004
Articles in Peer-reviewed Journals	Cerutti, F.;Ballarini, F.; Battistoni. G.; Cavinato, M.; Empl, A.; Fabrici, E.; Fassò, A.; Ferrari, A.; Gadioli, E.; Gadioli Erba, E.; Garzelli, M.V.; Ottolenghi, A.; Parini, V.; Pinsky, L.S.; Ranft, J.; Sala, P.R. "Towards a comprehensive description of heavy ion reactions" NATO Science Series, Kluwer Academic Publishers (in press) , Sep-2004
Articles in Peer-reviewed Journals	Andersen, V.; Ballarini, F.; Battistoni, F.; Cerutti, F.; Empl, A.; Fasso, A.; Ferrari, A.; Garzelli, M.V.; Ottolenghi, A.; Paretzke, H.; Pinsky, L.S.; Ranft, J.; Sala. P.; Wilson, T.; Zankl, M. "The Application of FLUKA to Dosimetry and Radiation Therapy" Radiation Protection Dosimetry (in press) , Sep-2004
Articles in Peer-reviewed Journals	Ferrari, A.; Pelliccioni, M.; Villari, R. "Evaluation of the influence of aircraft shielding on the aircrew exposure through an aircraft mathematical model" Radiat. Prot. Dosim. 108 (2), 91-105 (2004) , Sep-2004
Articles in Peer-reviewed Journals	Andersen, V.; Wilson, T. L.; Pinsky, L. S. "Helium production of prompt neutrinos on the Moon," Lunar and Planetary Science XXXV, 1809 , Mar-2004
Papers from Meeting Proceedings	Cerutti, F.; Ballarini, F.; Battistoni, G.; Cavinato, M.; Empl, A.; Fabrici, E.; Fassò, A.; Ferrari, A.; Gadioli, E.; Gadioli Erba, E.; Garzelli, M.; Ottolenghi; Parini, V.; Pinsky, L.S.; Ranft, J.; Sala, P. R "Toward a comprehensive description of heavy ion reactions" Structure and Dynamics of Elementary Matter, Proc. NATO ASI Conf. on Structure and Dynamics of Elementary Matter, Camyuva-Kemer (Antalya), Turkey Sep-2003
Papers from Meeting Proceedings	Andersen, V.; Ballarini, F.; Battistoni, G.; Campanella, M.; Carboni, M.; Cerutti, F.; Empl, A.; Fassò, A.; Ferrari, A.; Gadioli, E. "Progress towards a FLUKA based simulation tool aimed at the evaluation of space radiation environments" AIP Conference Proceedings 698, ed. Z. Parsa, 349-352 Feb-2004
Papers from Meeting Proceedings	Wilson, T.; Andersen, V.; and Pinsky, L. "Prompt Neutrino Production by the Lunar Surface" Proc. 28th International Cosmic Ray Conference, Tsukuba, Japan (Universal Academy Press), 1147-1150. (2003) Dec-2003
Presentation	Ballarina, F.; Battistoni, G.; Ceruti, F.; Empl, A.; Fasso, A.; Ferrari, A.; Gadioli, E.; Garzelli, M.; Ottolenghi, A.; Pinsky, L.; Ranft, J.; Roesler, S.; Sala, P.; and Scannicchio, D. "Nuclear models in FLUKA: a review" 10th Int. Conf. on Nuclear Reaction Mechanisms, Varenna, Italy Jun-2003
Presentation	Rollet, S.; Autischer, M.; Beck, P.; Ferrari, A.; M. Pelliccioni, M. "Dosimetric considerations on TEPC Fluka simulation and measurements" NEUDOS9, 9th Neutron Dosimetry Symposium Sep-2003
Presentation	Andersen, V.; Ballarini, F.; Battistoni, G.; Cerutti, F.; Empl, A.; Fasso, A.; Ferrari, A.; Garzelli, M.V.; Ottolenghi, A.; Paretzke, H.; Pinsky, L.; Ranft, J; Sala, P.; Wilson, T.L.; Zankl, M. "The application of FLUKA to dosimetry and radiation therapy" International Conference on Radiation Shielding (IRCS-10) and 13th Topical meeting on Radiation Protection and Shielding (RPS-2004), Funchal, Madeira Island, Portugal May-2004
Presentation	Pinsky, L.; Andersen, V.; Brun, R.; Cerutti, F.; Empl, A.; Fasso, A.; Ferrari, A.; Futo, E.; Garzelli, M.V.; Ranft, J; Sala, P.; Wilson, T.L.; Zapp, N "A FLUKA-based Monte Carlo transport code for space radiation simulation" International Conference on Radiation Shielding (IRCS-10) and 13th Topical meeting on Radiation Protection and Shielding (RPS-2004), Funchal, Madeira Island, Portugal May-2004
Presentation	Ballarini, F.; Cerutti, F.; Fassò, A.; Ferrari, A.; Gadioli, E.; Garzelli, M.V.; Ottolenghi, A.; Pinsky, L.S.; Ranft, J.; Sala, P.R "Heavy-ion collisions in the FLUKA Monte Carlo event generator: present and perspective" 2004 International Nuclear Physics Conference (2004 INPC), Goteborg, Sweden Jun-2004
Presentation	Ballarini, F.; Cerutti, F.; Ferrari, A.; Gadioli, E.; Garzelli, M.V.; Ottolenghi, A.; Paretzke, H.; Parini, V.; Pelliccioni, M.; Pinsky, L.; Sala, P.; Zankl, M "Models and Monte Carlo simulations of GCR and SPE organ doses with different shielding, based on the FLUKA code coupled with anthropomorphic phantoms" 35th COSPAR Scientific Assembly, Paris, France Jul-2004
Presentation	Ferrari, A.; Ballarini, F.; Battistoni, G.; Cerutti, F.; Empl, A.; Fassò, A.;Gadioli, E.; Garzelli, M.V.; Ottolenghi, A.; Pinsky, L.; Ranft, J.; Sala, P.R.; Smirnov, G.; Roesler, S.; Zapp, N "The FLUKA code: new developments and applications for space radiation" 35th COSPAR Scientific Assembly, Paris, France Jul-2004
Presentation	Lee, K.; Pinsky, L.; Andersen, V.; Zeitlin, C.; Cleghorn, T.; Cucinotta, F.; Saganti, P.; Atwell, W.; Turner, R "Helium Cosmic Ray Measurements at Mars" 35th COSPAR Scientific Assembly, Paris, France Jul-2004
Presentation	Ranft, J.; Bopp, F.W.; Engel, R.; Roesler, S. "Learning from RHIC data for the DPMJET model" Quark Matter 2004, Oakland, CA Jan-2004
Presentation	Pinsky, L. "NASA's Interest in Cosmic Rays: Radiation Hazards to a Human Presence in Space, A Brief Introduction, (Invited Talk)" Vulcano Cosmic Ray Workshop, Vulcano Island, Sicily May-2004

Presentation	Pinsky, L.; Anderson, V.; Empl, A.; Lee, K.; Smirnov, G.; Zapp, N.; Ferrari, A.; Tsoulou, K.; Roesler, S.; Vlachoudis, V.; Battistoni, G.; Cerutti, F.; Gadioli, E.; Garzelli, M.V.; Muraro, S.; Rancati, T.; Sala, P.; Ballarini, F.; Ottolenghi, A.; Parini, V.; Scannicchio, D.; Pelliccioni, M.; Wilson, T.L.; Ranft, J.; Fasso, A "Update On the Status of the FLUKA Monte Carlo Transport Code" Computing In High Energy Physics (CHEP 04), Interlaken, Switzerland Sep-2004
Presentation	Battistoni, G.; Ferrari, A.; Tsoulou, K.; Roesler, S.; Vlachoudis, V.; Cerutti, F.; Gadioli, E.; Garzelli, M.V.; Muraro, S.; Rancati, T.; Sala, P.; Ballarini, F.; Ottolenghi, A.; Parini, V.; Scannicchio, D.; Pelliccioni, M.; Anderson, V.; Empl, A.; Lee, K.; Pinsky, L.; Smirnov, G.; Zapp, N.; Wilson, T.L.; Ranft, J.; Fasso, A. "Application of the FLUKA Monte Carlo Code in Accelerator and High Energy Physics" Computing In High Energy Physics (CHEP 04), Interlaken, Switzerland Sep-2004
Presentation	Pinsky, L.S "The Status of Radiation Science at the University of Houston, (Invited Talk)" M.D. Anderson Cancer Center, Houston, TX Sep-2004
Presentation	Townsend, L. W "Implications of the Space Radiation Environment for Human Exploration in Deep Space (Plenary Presentation)" 10th International Conference on Radiation Shielding/13th ANS Topical Meeting on Radiation Protection and Shielding, Funchal, Madeira, Portugal May-2004
Presentation	Townsend, L. W.; Miller, T. M.; and Gabriel, T. A. ".: HETC Radiation Transport Code Development for Cosmic Ray Shielding Applications in Space" 10th International Conference on Radiation Shielding/13th ANS Topical Meeting on Radiation Protection and Shielding, Funchal, Portugal May-2004
Presentation	Miller, T. M.; Townsend, L. W.; Gabriel, T. A.; Handler, T "Extension of the HETC Radiation Transport Code to Include HZE Particle Transport" 3rd International Workshop on Space Radiation Research (IWSRR) and 15th Space Radiation Health Investigator's Workshop, Port Jefferson, NY May-2004
Presentation	Miller, T. M. and Townsend, L. W. "Double-Differential Light Ion Production Cross Sections" Ninth Symposium on Neutron Dosimetry, NEUDOS 9, Delft, The Netherlands Sep-2003
Presentation	Miller, T. M. and Townsend, L. W. "Double-Differential Heavy Ion Production Cross Sections" Ninth Symposium on Neutron Dosimetry, NEUDOS 9, Delft, The Netherlands Sep-2003
Presentation	Townsend, L. W. "NASA Space Radiation Transport Code Development Consortium" 10th International Conference on Radiation Shielding/13th ANS Topical Meeting on Radiation Protection and Shielding, Funchal, Portugal May-2004
Presentation	Zapp, N.; Garzelli, M.-V.; Empl, A.; Ferrari, A.; Wilson, T.; Townsend, L.; Pinsky, L. "HMD Development: An Event Generator for Monte Carlo Simulation of Heavy Ion Transport" . 3rd International Workshop on Space Radiation Research (IWSRR) and 15th Space Radiation Health Investigator's Workshop, Port Jefferson, NY May-2004
Presentation	Miller, T. M.; Townsend, L. W.; Gabriel, T. A.; Handler, T. ".: Comparison of Secondary Fluence Predictions of the Radiation Transport Code HETC-HEDS with High-Energy Heavy Ion Beam Laboratory Data" 2004 International Nuclear Physics Conference (2004 INPC), Goteborg, Sweden Jun-2004